

NSERC CREATE Grant Proposal

Project Title: Advanced Neuroimaging Registration Platform Using Finite Mathematics and Machine Learning

1. Executive Summary

This proposal seeks NSERC CREATE funding to support the development and validation of a novel neuroimaging registration application. The platform leverages finite mathematics, spherical harmonics, Gaussian Mixture Models (GMM), and continued fractions for robust, verifiable, and efficient diffeomorphic registration of brain surfaces. The project will provide advanced tools for the Canadian neuroscience and medical imaging communities, with a focus on formal mathematical verifiability and ground truth validation.

2. Introduction and Rationale

Accurate registration of neuroimaging data is critical for research and clinical applications. Current methods often lack formal guarantees of correctness and struggle with complex anatomical variability. Our app, developed in Python with a modern web interface, addresses these challenges by integrating:

- Spherical harmonics and Legendre polynomials for surface representation
- GMM-based deformable registration
- Continued fraction-based registration
- Formal verifiability of diffeomorphic mappings
- Ground truth validation using synthetic and real datasets

3. Objectives

1. Develop a user-friendly, web-based neuroimaging registration tool.
2. Implement finite math-based surface modeling (spherical harmonics, Legendre polynomials).
3. Integrate GMM and continued fraction registration algorithms.
4. Provide formal proofs and computational verification of diffeomorphic mappings.
5. Validate registration accuracy against ground truth datasets.

4. Methodology

4.1 Finite Mathematics for Surface Modeling

Spherical harmonics: $Y_l^m(\theta, \phi) = \sqrt{(2l+1)/(4\pi) * (l-m)!/(l+m)!} * P_l^m(\cos\theta) * \exp(im\phi)$, where P_l^m are the associated Legendre polynomials.

4.2 Gaussian Mixture Model Registration

Given source points X and target points Y , registration minimizes: $L(\theta) = -\sum_i \log \sum_k \pi_k N(y_i | \mu_k, \Sigma_k)$, where N is the Gaussian density, π_k are mixture weights, and θ are model parameters.

4.3 Continued Fraction Registration

We use continued fractions to model non-linear deformations:

$x = a_0 + 1/(a_1 + 1/(a_2 + 1/(a_3 + \dots)))$

This allows flexible, invertible mappings for surface registration.

4.4 Formal Verifiability

We provide computational proofs that the registration mapping $f: S^2 \rightarrow S^2$ is diffeomorphic, i.e., f is smooth and invertible with a smooth inverse. This is verified by checking the Jacobian determinant $\det J_f > 0$ everywhere on the surface.

4.5 Ground Truth Validation

Validation is performed using synthetic datasets with known transformations and real DICOM data. Registration error is quantified as:

$$TRE = (1/N) \sum_i || f(x_i) - y_i ||$$

5. App Snapshots

[Insert: finite_math_stability.png]

Figure 1: Finite math stability visualization.

[Insert: mesh_superimpose.png]

Figure 2: Mesh registration overlay.

[Insert: continued_fraction_convergence.png]

Figure 3: Continued fraction registration convergence.

6. Training and HQP Development

The project will train graduate students and postdocs in advanced mathematics, machine learning, and software engineering. Workshops and hackathons will be organized.

7. Timeline and Milestones

- Year 1: Core algorithm development, app prototype
- Year 2: Formal verification, user testing, ground truth validation
- Year 3: Dissemination, training, and open-source release

8. Budget and Justification

- Personnel: \$120,000
- Equipment: \$30,000
- Software & Cloud: \$20,000
- Workshops: \$10,000
- Total: \$180,000

9. References

1. Bookstein, F.L. (1997). Morphometric Tools for Landmark Data.
2. Ashburner, J., & Friston, K.J. (1999). Nonlinear spatial normalization using basis functions.
3. Durrleman, S. et al. (2014). Morphometry of anatomical shape complexes.

End of Proposal